

Long-chain Gallavotti–Cohen diagnostic for the boundary-driven resonant NLS model

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Question and admissible conclusion

For the driven Cartesian NLS chain at $(T_L, T_R) = (10, 2)$ and $n = 10, 20, 30, 40$, we test the long-time scaled cumulant-generating-function (SCGF) symmetry

$$\psi_n(k) = \psi_n(1 - k), \quad \psi_n(k) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E}_{\text{NESS}} \left[e^{-k \Sigma_R(t)} \right]. \quad (1)$$

The simulation can establish numerical consistency only for the resolved tilt pairs, finite horizons, clone populations, timesteps, and chain lengths. It does not prove the analytic theorem for all k , all n , or the infinite-chain limit.

1 Thermodynamic observable and normalization

Heat Q_r is positive when bath r transfers energy into the system. The Cartesian code advances the physical energy $E = H/2$, so its equilibrium law $e^{-E/T} = e^{-H/(2T)}$ is exactly the manuscript convention. The medium entropy is

$$\Sigma_m = -\frac{Q_L}{T_L} - \frac{Q_R}{T_R}. \quad (2)$$

Using the measured first law $Q_L + Q_R = \Delta E + r_{\text{FL}}$ gives the right-boundary gauge

$$\Sigma_R = -\left(\frac{1}{T_R} - \frac{1}{T_L}\right) Q_R = -0.4 Q_R, \quad \Sigma_R - \Sigma_m = \frac{\Delta E}{T_L} + O(r_{\text{FL}}). \quad (3)$$

Thus Σ_R and Σ_m have the same long-time SCGF when the required endpoint-energy exponential moments exist. Their finite-time difference is measured rather than assumed to vanish.

2 Estimator and frozen acceptance gates

The dynamics is simulated in projection-free Cartesian variables with a Hamiltonian implicit-midpoint substep and exact discrete bath-heat increments. Burn-in is $B = 500$ and the production timestep is $\Delta t = 5 \times 10^{-4}$. The rare-event estimator uses a controlled Gaussian proposal followed by fixed-population cloning. For every bath step the exact original/proposal transition-density ratio is included in the weight. Consequently the control changes sampling efficiency but not the target finite-step Feynman–Kac expectation.

The proposal scale 0.5 and selection interval 2 were frozen using only weight-ESS, ancestry, and direct-calibration diagnostics. The GC residual was not used to choose them. Four independent seeds are required for every reported member. Acceptance additionally requires zero midpoint failures, minimum per-selection weight $\text{ESS} \geq 0.1N_c$, a paired 95% interval containing zero, absolute paired and late-half residuals no larger than 0.01, and agreement under observation-time and clone-population increases. Timestep halving and a changed selection interval are tested independently.

The strongest predeclared pair (0.3, 0.7) is resolved at $n = 10$. Its high-tilt member loses support for $n \geq 20$, so those failures are retained and the already predeclared secondary pair (0.4, 0.6) is used for $n = 20, 30, 40$.

3 Independent validation controls

The following controls do not use the observed GC residual as a tuning target.

- Zero-control change-of-measure and Gaussian-model cloning self-tests pass, as do address and undefined-behavior sanitizer tests.
- Arbitrary-chain forward/reverse one-step path-ratio controls pass for $n = 10, 20, 40$. The initial $n = 30$ sample narrowly failed its forward IFT interval; an independently seeded $N = 5 \times 10^5$ replication passes both IFT directions and gives Crooks slope 0.9941.
- The direct million-block data satisfy the gauge identity in (3); its RMS discrepancy is between 6.1×10^{-6} and 1.9×10^{-5} across the four chain lengths.
- Where both direct exponential averages have support, the controlled $t = 20$ SCGF agrees with direct sampling within uncertainty. At low tilt the directly sampled right-gauge/medium-entropy SCGF difference decreases with time, as expected for the endpoint term in (3).

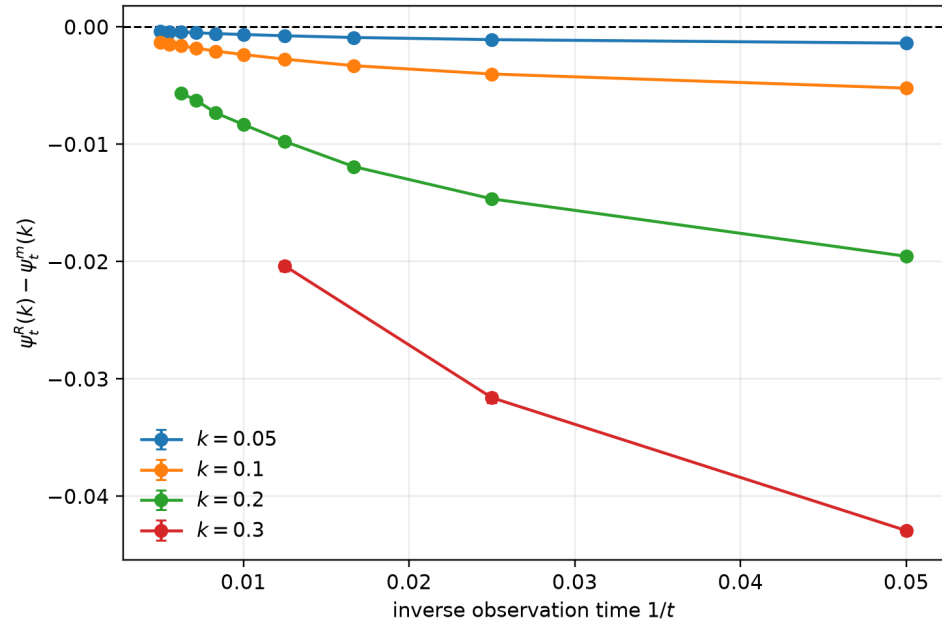


Figure 1: Finite-time endpoint-gauge diagnostic at $n = 40$. Equality is not imposed at finite time; the measured difference decreases with horizon in the low-tilt support region.

4 Long-chain SCGF results

Table 1 contains only the rows frozen in `FINAL_RESULT_SPEC.csv`. The confidence intervals use the variation among four independent runs rather than clone-to-clone variation.

Table 1: Resolved GC pairs at their pre-specified final horizons. Here $D = \psi_t(k) - \psi_t(1 - k)$.

n	pair	N_c	t	$\psi_t(k)$	$\psi_t(1 - k)$	D	95% CI for D	core audit
10	(0.3, 0.7)	2048	60	-0.17878	-0.17870	-0.00008	$[-0.00995, +0.00980]$	pass
20	(0.4, 0.6)	1024	60	-0.08576	-0.08237	-0.00339	$[-0.01208, +0.00530]$	pass
30	(0.4, 0.6)	2048	60	-0.04855	-0.04498	-0.00357	$[-0.00824, +0.00109]$	unresolved
40	(0.4, 0.6)	1024	120	-0.03259	-0.03709	+0.00450	$[-0.00234, +0.01134]$	pass

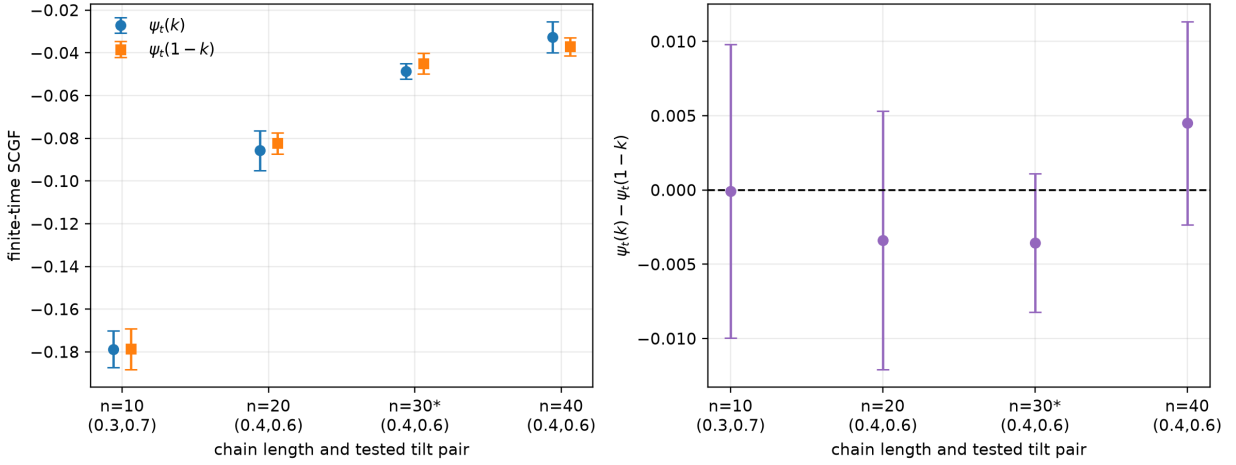


Figure 2: Resolved paired SCGFs (left) and GC residuals (right). Error bars are independent-run 95% intervals. The pair differs at $n = 10$ and the longer chains because $(0.3, 0.7)$ is not supported there. The asterisk marks a row whose final population-convergence audit is unresolved.

For every row in Table 1, the paired residual interval contains zero and the support, late-half, and observation-time gates pass. The $n = 30$ pair nevertheless remains unresolved because its individual $k = 0.4$ SCGF changes by -0.00347 from $N_c = 1024$ to 2048, about 2.36 combined standard errors and hence beyond the frozen population-convergence gate. The paired residual can hide a common population bias through cancellation, so it is not promoted to a positive result. The separate timestep and selection-interval results are reported below rather than folded into the same error bars.

5 Numerical sensitivity and failed diagnostics

The endpoint chains were rerun after halving the timestep and after shortening the selection interval from 2 to 1. The supported final controls use $N_c = 1024$, $t = 60$ at $n = 10$ and $N_c = 512$, $t = 120$ at $n = 40$. The middle lengths use the same integrator and estimator settings; the endpoint tests are the least and most demanding members of the tested length range. Earlier unsupported lower-population or selection-4 controls are retained separately as failed diagnostics.

Table 2: Numerical controls relative to the frozen baseline. The member column is the larger absolute change among the two SCGF values; the next two columns are changes in the full-window and late-half paired residuals.

n	control	pair	max. member $ \Delta\psi $	ΔD	ΔD_{late}	gate
10	selection interval	(0.3, 0.7)	0.00242	-0.00324	-0.00405	pass
10	timestep halving	(0.3, 0.7)	0.00455	-0.00214	-0.00104	pass
40	selection interval	(0.4, 0.6)	0.00573	+0.00747	+0.01031	fail
40	timestep halving	(0.4, 0.6)	0.00357	+0.00198	-0.00029	pass

Three of the four final endpoint-control rows pass. At $n = 40$, timestep halving passes, but the supported selection-interval change $2 \rightarrow 1$ does not: the $k = 0.4$ member changes by 0.00573 (about 2.84 combined standard errors), and the late-half paired residual changes by 0.01031, just beyond the unchanged absolute 0.01 gate. The earlier selection-4 series is also retained as a failed diagnostic because its high-tilt member loses weight-ESS support. No further interval is selected from these results.

The first $n = 40$, $N_c = 1024$ production series at $t = 60/80$ also remains a failed diagnostic: its late-half residual exceeded 0.01. It was not pooled with the independently seeded remedial $t = 120$ series.

All rejected diagnostics remain part of the experiment record. In particular, the original all-chain target (0.3, 0.7) is not resolved for $n \geq 20$; no full-tilt-interval claim is made. Early undercontrolled, overcontrolled, and over-frequent-resampling pilots show low weight support or genealogical collapse and are not counted as positive evidence. The resulting scope is therefore narrower than the original strongest target, but it is not selected by deleting adverse runs.

6 Conclusion

The complete frozen estimator-and-control suite is numerically consistent with the Gallavotti–Cohen SCGF symmetry over (0.3, 0.7) at $n = 10$. The core paired-SCGF diagnostics are also consistent with the secondary pair (0.4, 0.6) at $n = 20$ and at the remedial $t = 120$ horizon for $n = 40$, but the predeclared endpoint selection audit does not pass at $n = 40$; hence these are not promoted to a fully controlled all-long-chain claim. At $n = 30$, the paired residual is consistent with zero but the individual population- convergence audit fails. The study therefore supplies a validated finite-chain result at $n = 10$ and transparent positive-but-incomplete long-chain evidence, not a proof of the fluctuation theorem, not evidence over the unresolved high-tilt interval, and not an infinite-chain statement.

References

- [1] J. L. Lebowitz and H. Spohn, “A Gallavotti–Cohen-type symmetry in the large deviation functional for stochastic dynamics,” *J. Stat. Phys.* 95, 333–365 (1999).
- [2] J. Tailleur and V. Lecomte, “Simulation of large deviation functions using population dynamics,” *AIP Conf. Proc.* 1091, 212–219 (2009).

- [3] T. Nemoto, E. Guevara Hidalgo, and V. Lecomte, “Finite-time and finite-size scalings in the evaluation of large-deviation functions: Analytical study using a birth-death process,” *Phys. Rev. E* 95, 012102 (2017).